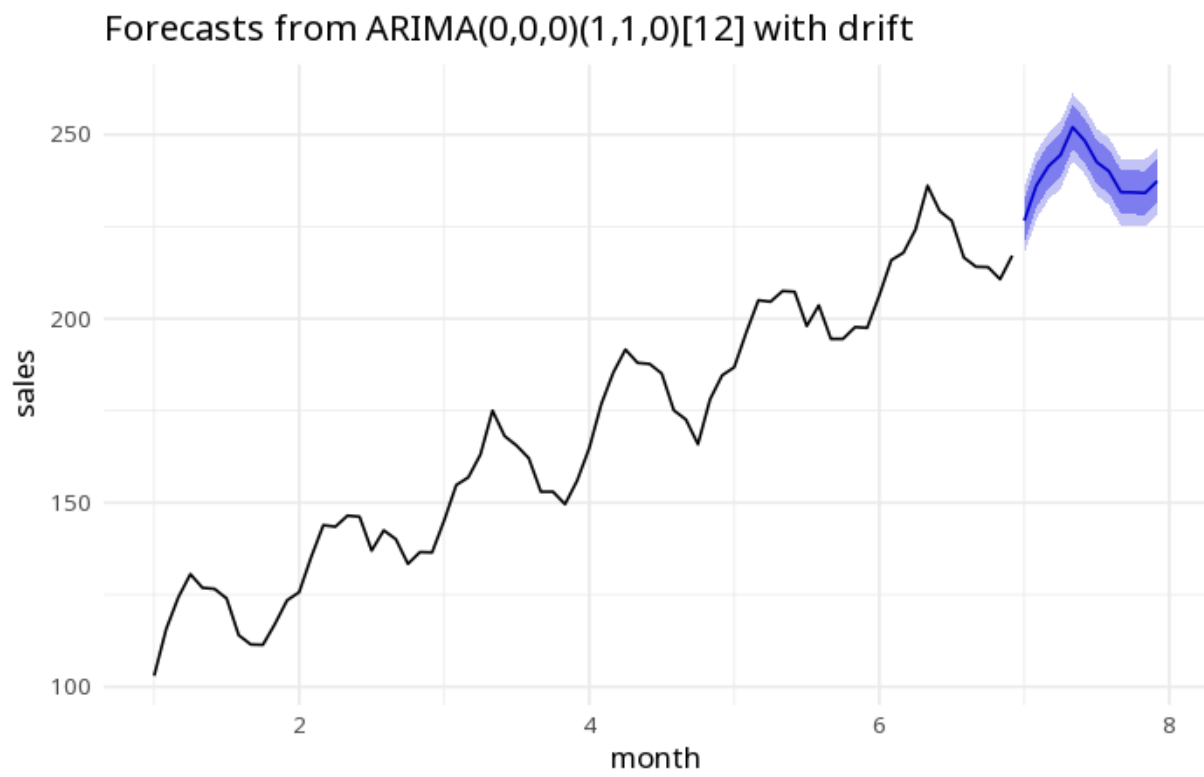
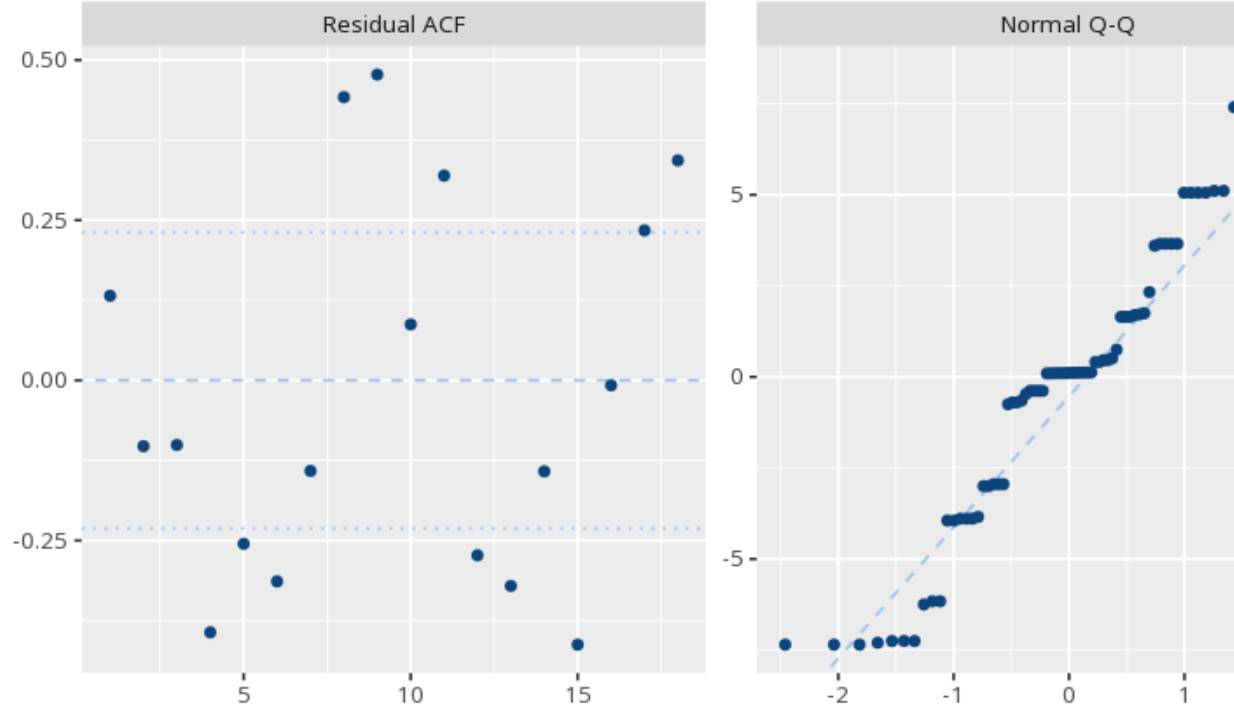


1 ARIMA / SARIMA

(1)			
sar1	−0.48 (0.12) [−0.71, −0.25]		
drift	1.67 (0.04) [1.60, 1.74]		
Num.Obs.	72		
σ^2	21.66		
Ljung–Box p	<.001		
AIC	361.95		
BIC	368.23		
Log.Lik.	−177.97		
Ljung–Box Q(23) = 169.43, p < .001 (lag 24, residual autocorrelation)			
Period	Forecast	80% PI	95% PI
1	226.60	[220.64, 232.56]	[217.48, 235.72]
2	236.15	[230.19, 242.12]	[227.03, 245.27]
3	241.39	[235.43, 247.36]	[232.27, 250.51]
4	244.45	[238.49, 250.42]	[235.33, 253.57]
5	252.00	[246.03, 257.96]	[242.88, 261.12]
6	248.34	[242.38, 254.30]	[239.22, 257.46]
7	242.50	[236.53, 248.46]	[233.38, 251.62]
8	240.04	[234.08, 246.01]	[230.92, 249.17]
9	234.35	[228.39, 240.32]	[225.23, 243.47]
10	234.30	[228.34, 240.26]	[225.18, 243.42]
11	234.14	[228.18, 240.11]	[225.02, 243.27]
12	237.35	[231.39, 243.32]	[228.23, 246.47]





The chosen $(p,d,q)(P,D,Q)$ orders describe the model. The Ljung–Box $Q(df)$ row tests whether the residuals still carry autocorrelation up to the reported lag (a large p means none is detected); use it with AIC/BIC to judge fit. The forecast table and plot give predictions with uncertainty bands. AIC/BIC compare models only when fit to the same series with identical differencing (d, D) and transformation; lower is better, but they are relative, not absolute, measures of fit. Method: R forecast package (Hyndman & Khandakar, 2008; Hyndman & Athanasopoulos, Forecasting: Principles and Practice).

An $ARIMA(0,0,0)(1,1,0[12])$ model was fit ($AIC=361.95$); the Ljung–Box test of residual autocorrelation gave $p < .001$.

Statistical basis: ARIMA/SARIMA model selection and forecasting (Hyndman R, Athanasopoulos G, Bergmeir C, et al. (2024). forecast: Forecasting functions for time series and linear models. R package.).